Statistical Foundations of Learning

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1 Statistical learning problem and Bayes risk

1.1 Learning problem and risk

Setup. Inputs live in a **feature space** $\mathcal{X}$, outputs in a **label set** $\mathcal{Y}$ (binary classification: $\mathcal{Y} = \{0, 1\}$ or $\{\pm 1\}$; regression: $\mathcal{Y} \subseteq \mathbb{R}$). A **predictor** is a map $h : \mathcal{X} \to \mathcal{Y}$. Given a **training sample** $S = ((x_i, y_i))_{i=1}^m \in (\mathcal{X} \times \mathcal{Y})^m$, a **learner** $\mathcal{A}$ returns a predictor $\hat{h}_S = \mathcal{A}(S)$; it must accept samples of every size $m \geq 1$ (think of a neural network trained by gradient descent on S).

Data model. Learning only makes sense if test data resemble training data. We assume a fixed but unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$: training and test pairs are all drawn from $\mathcal{D}$, independently, so $S \sim \mathcal{D}^m$. Equivalently, $\mathcal{D}$ is specified by the marginal $\mathcal{D}_\mathcal{X}$ of x and the conditional law $\mathbb{P}(y \mid x)$ of the label. A deterministic labelling $y = f(x)$ is the special case of a degenerate conditional; it cannot express that two equal inputs may carry different labels.

Loss and risk. A **loss** $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$ penalises a prediction $h(x)$ when the truth is y. The two standard choices are the **0-1 loss** $\mathbb{1}\{h(x) \neq y\}$ for classification and the **squared loss** $(h(x) - y)^2$ for regression.

Definition 1.1 (Risk)**.** The **risk** (generalisation error) of h is its average loss on a new pair from $\mathcal{D}$,

$$L_\mathcal{D}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h(x), y)\big].$$

For a learner, the quantity of interest is $L_\mathcal{D}(\hat{h}_S)$, the expected test error of the predictor learned from S.

Under the 0-1 loss, $L_\mathcal{D}(h)$ is the probability that h misclassifies, $L_\mathcal{D}(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(h(x) \neq y)$.

Example 1.2 (Two Gaussian classes)**.** Take $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{\pm 1\}$, both labels with probability $\frac{1}{2}$, and class-conditional laws $\mathcal{N}(-\mu, \sigma^2)$ for $y = -1$ and $\mathcal{N}(\mu, \sigma^2)$ for $y = +1$, with $\mu > 0$. Then $\mathcal{D}_\mathcal{X} = \frac{1}{2}\mathcal{N}(-\mu, \sigma^2) + \frac{1}{2}\mathcal{N}(\mu, \sigma^2)$, and $\mathbb{P}(y = +1 \mid x)$ increases from 0 to 1, crossing $\frac{1}{2}$ at $x = 0$. A threshold rule $h(x) = \operatorname{sign}(x - t)$ errs on the $+1$ points left of t and on the -1 points right of t (fig. 1); moving t away from 0 trades errors on one class for errors on the other. The optimal threshold is $t = 0$, where $\operatorname{sign}(x)$ always predicts the more likely label (theorem 1.6); in practice it has to be found from S alone.

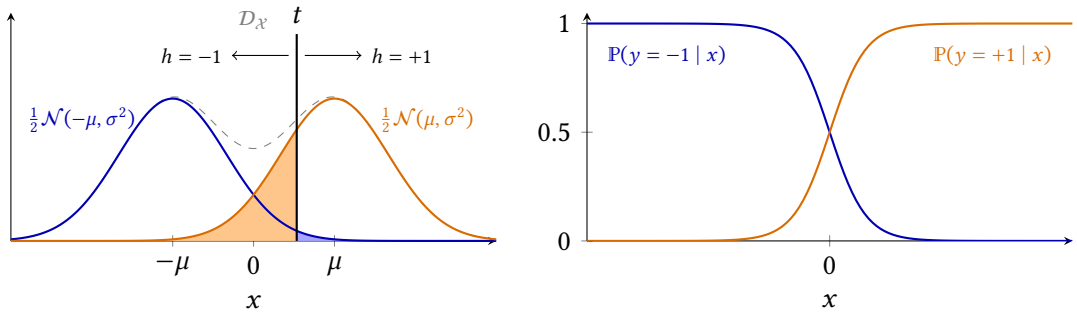


Figure 1: Left: weighted class densities, $\mathcal{D}_\mathcal{X}$ (dashed), errors of $\operatorname{sign}(x - t)$ (shaded). Right: $\mathbb{P}(y = \pm 1 \mid x)$.

1.2 Empirical risk minimisation

Ideally we would solve $\min_{h \in \mathcal{Y}^\mathcal{X}} L_\mathcal{D}(h)$, but $L_\mathcal{D}(h)$ cannot be evaluated without knowing $\mathcal{D}$, so it is replaced by its sample counterpart.

Definition 1.3 (Empirical risk, ERM)**.** The **empirical risk** (training error) of h on S is $L_S(h) = \frac{1}{m}\sum_{i=1}^m \ell(h(x_i), y_i)$. **Empirical risk minimisation** (ERM) solves $\min_{h \in \mathcal{Y}^\mathcal{X}} L_S(h)$.

For fixed h, $L_S(h)$ is an unbiased estimate of $L_D(h)$: $\mathbb{E}_{S\sim D^m}[L_S(h)] = L_D(h)$. Minimising it over *all* functions, however, can overfit badly:

Example 1.4 (ERM can fail)**.** Let $x \sim \text{Uniform}([0,1])$ and $y = \mathbb{1}\{x > 0.1\}$. The predictor that outputs 1 on the training points labelled 1 and 0 everywhere else has $L_S(\hat{h}) = 0$. Off the finite set $\{x_1, \dots, x_m\}$ it always says 0, so $L_D(\hat{h}) = \mathbb{P}(x > 0.1) = 0.9$.

1.3 Bayes risk

Suppose now that D is known. The **Bayes risk** is $L_D^* = \min_{h\in\mathcal{Y}^{\mathcal{X}}} L_D(h)$, and a minimiser h^* is a **Bayes predictor**.

Classification. Let $\mathcal{Y} = \{0,1\}$, ℓ the 0-1 loss, and write $\eta(x) = \mathbb{P}(y = 1 \mid x)$, so that $\mathbb{P}(y = 0 \mid x) = 1 - \eta(x)$.

Theorem 1.5 (Risk of a binary classifier)**.** *For every deterministic $h : \mathcal{X} \to \{0,1\}$,*

$$L_D(h) = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\mathbb{P}(h(x) \neq y \mid x)\big] = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}\big].$$

Proof. Conditioning on x gives the first equality. Then split $\mathbb{1}\{h(x) \neq y\} = \mathbb{1}\{h(x) = 0\}\mathbb{1}\{y = 1\} + \mathbb{1}\{h(x) = 1\}\mathbb{1}\{y = 0\}$: because h is deterministic, $h(x)$ is a constant once x is fixed, so only the indicators of y are averaged, giving $\eta(x)$ and $1 - \eta(x)$. □

At each x, a classifier therefore pays $\eta(x)$ by predicting 0 or $1 - \eta(x)$ by predicting 1. The best choice is the cheaper option, which defines the **Bayes classifier**

$$h^*(x) = \mathbb{1}\{\eta(x) \geq \tfrac{1}{2}\}.$$

Theorem 1.6 (Bayes risk equals the risk of the Bayes classifier)**.** *For the 0-1 loss,*

$$L_D^* = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\min\{\eta(x), 1 - \eta(x)\}\big] = L_D(h^*) = \min_{h\in\{0,1\}^{\mathcal{X}}} L_D(h).$$

Proof. In theorem 1.5 the integrand is at least $\min\{\eta(x), 1-\eta(x)\}$ for any h, which gives the lower bound. For $h = h^*$ it equals that minimum at every x (ties at $\eta = \frac{1}{2}$ cost the same either way). □

Example 1.7 (Installation crash)**.** Let $x \sim \text{Uniform}[0,1]$ be the completed fraction of a software installation, with $y = 1$ meaning that the system crashes after the fraction x has been completed, and $\eta(x) = |1 - 2x|$. Then $\eta(x) \geq \frac{1}{2}$ iff $x \leq \frac{1}{4}$ or $x \geq \frac{3}{4}$, so $h^* = 1$ on $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$ and $h^* = 0$ on $(\frac{1}{4}, \frac{3}{4})$, with $L_D^* = L_D(h^*) = \frac{1}{4}$.

Regression. Let $\mathcal{Y} = \mathbb{R}$ with the squared loss, so that $L_D(h) = \mathbb{E}[(h(x) - y)^2]$. ERM is then least-squares regression (ordinary least squares when h is restricted to linear functions).

Proposition 1.8 (Bayes regressor)**.** *For the squared loss, $h^*(x) = \mathbb{E}[y \mid x]$ and $L_D^* = \mathbb{E}_x\big[\operatorname{Var}(y \mid x)\big]$.*

Proof. Expanding the square and conditioning on x, for any deterministic h,

$$L_D(h) = \mathbb{E}_x\big[(h(x) - \mathbb{E}[y \mid x])^2\big] + \mathbb{E}_x\big[\mathbb{E}[y^2 \mid x] - (\mathbb{E}[y \mid x])^2\big].$$

The second term is $\mathbb{E}_x[\operatorname{Var}(y \mid x)]$ and does not involve h; the first is non-negative and vanishes for $h(x) = \mathbb{E}[y \mid x]$. □

Example 1.9 (Signal plus noise)**.** If $y = f(x) + \epsilon$ with ϵ independent of x, $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon^2] = \sigma^2$, then $\mathbb{E}[y \mid x] = f(x)$ and $\operatorname{Var}(y \mid x) = \sigma^2$, so $h^* = f$ and $L_D^* = \sigma^2$.